

1.1 m.s. speed. Thus, some helium particles that have **with speed greater than the escape speed will escape** the Earth's atmosphere. [B1]

4 (a)

free oscillation: oscillation in which a body oscillates without any loss of energy or without resistive forces or without external forces applied. [B1]

forced oscillation: oscillation in which a body is made to vibrate by an external periodic force or continuous energy is transferred to the vibrating body. [B1]

4 (b)(i)

Resonance can be used as the driving frequency of the oscillator matches the natural frequency of the vibration of the trolley. [B1]

It can be observed where the amplitude of the vibration is the maximum. [B1]

From Fig. 4.2, frequency is 2.1 Hz (allow 2.08 to 2.12 Hz).

[B1]

4 (b)(ii)

The peak of the graph is not very sharp or the amplitude of the vibration is finite, so frictional forces are present. [B1]

4 (c)

$$V_0 = \omega x_0$$

$$= 2\pi \times 2.1 \times 4.7 \times 10^{-2}$$

[C1]

$$= 0.62 \text{ m s}^{-1}$$

[A1]

5 (a)